

THE THREE TRADITIONS OF EMBODIED INTELLIGENCE

Synthesizing Machine Learning (Brain), Real-Time Embedded (Nervous System), and Robotics (Body)

1. Machine Learning

Computer Science · The Brain

Core Superpower:

Open-world semantic competence, visual foundation models, high-dimensional representation learning.

Critical Blindspot:

Stochastic outputs, unbounded tail latency, zero deterministic physical safety guarantees.

Key Primitives:

Transformers · Diffusion · Statistical Loss

2. Embedded Systems

ECE / Systems · Nervous System

Core Superpower:

Microsecond clock determinism, bare-metal register control, zero-allocation SRAM, fault tolerance.

Critical Blindspot:

Minimal semantic capacity, cannot generalize to unstructured open worlds.

Key Primitives:

1 kHz Timers · DMA Seqlocks · Watchdogs

3. Robotics Dynamics

Mechanical Eng · The Body

Core Superpower:

Analytical differential equations, Lyapunov stability envelopes, contact mechanics, conservation laws.

Critical Blindspot:

Requires precise analytical models; brittle under unmodeled real-world visual shifts.

Key Primitives:

Momentum $p = mv$ · Inertia · Heat Capacity

PHYSICAL ARTIFICIAL INTELLIGENCE · The Systems Engineering Synthesis

High-Capacity Stochastic Deliberation (**Brain**) Gated by Bare-Metal Safety Invariants (**Nervous System**) over Dynamical Physics (**Body**)